

# Knesset Protocol Journal Project

## Data Overview & Preprocessing Report

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### Dataset Overview

The raw dataset consists of **213 JSON files**, each representing a single Israeli Knesset (25th) Finance Committee (ועדת הכספים) meeting. Files follow the naming convention `25_ptv_<doc_id>.json`.

### Per-file Schema

| Field                    | Type     | Description                        |
|--------------------------|----------|------------------------------------|
| <code>knesset_num</code> | int      | Knesset session number (25)        |
| <code>committee</code>   | string   | Committee name (Hebrew)            |
| <code>doc_id</code>      | string   | Unique protocol identifier         |
| <code>date</code>        | ISO 8601 | Meeting date and time              |
| <code>source_file</code> | URL      | Source document on Knesset website |
| <code>text</code>        | string   | Full meeting transcript (Hebrew)   |

**Language:** All transcript text is in Hebrew. **Date range:** November 2022 – June 2023.

### Topic Header Extraction

**Primary Marker:** <<      >>

The vast majority of protocols embed topic headers using the pattern:

<<      >> *topic text* <<      >>

Each topic appears twice per file (once in the agenda block, once as a section header in the transcript body). We extract unique spans, deduplicate within file, and strip leading list numbering (e.g. 1. , 2. ) from multi-topic agendas.

**Fallback 1:** <<      >> (Budget Law Variant)

Four protocols use an alternate marker <<      >> instead of <<      >>. These are sessions dedicated to the 2023–2024 State Budget Law, which uses a different typographical convention in the official documents. We apply identical extraction logic to this marker.

**Fallback 2:** סדר היום Agenda Block

Four protocols carry no inline markers at all. For these, we extract the topic from the סדר היום (“agenda”) block at the top of every transcript, delimited between the meeting header and the נכחו (“attendees”) section.

## Extraction Results

| Method            | Files      | %           |
|-------------------|------------|-------------|
| << >> marker      | 205        | 96.2%       |
| << >> marker      | 4          | 1.9%        |
| סדר היום fallback | 4          | 1.9%        |
| No topic found    | 0          | 0.0%        |
| <b>Total</b>      | <b>213</b> | <b>100%</b> |

All 213 protocols were successfully assigned at least one topic.

## Topic Canonicalization

### Raw vs. Canonical

Raw extraction yielded **180 unique topic strings** across all 213 files. Many are near-duplicates caused by:

- Typographical errors (מיסים vs. מסים; סגירה vs. סגירת; להתקשרות vs. להתקשורת)
- Punctuation/dash variants (em-dash vs. regular hyphen)
- Bill number formatting ((1577/ ) vs. , 1577/ vs. ---2022, 1577/ )
- Missing הצעת prefix on otherwise identical strings
- Quote mark and whitespace variants (על ידי vs. על-ידי)

### Fuzzy Clustering Algorithm

We applied fuzzy string matching (`rapidfuzz` token ratio, threshold = 88/100) to cluster near-duplicates and assign a canonical representative per cluster (the longest string in each cluster — typically the most complete form).

The algorithm identified **23 non-trivial clusters**. After canonicalization, the vocabulary reduced from 180 raw strings to **151 canonical topics**.

### Manual Corrections

At threshold 88, fuzzy matching produced **14 false-positive merges** — cases where two genuinely distinct topics scored above the similarity threshold due to shared legal boilerplate phrasing. These were manually identified and corrected:

| Incorrectly merged pair                                                                        | Reason for split                  |
|------------------------------------------------------------------------------------------------|-----------------------------------|
| Budget amendments 2022 & 2023                                                                  | Different fiscal years            |
| State Budget Law 2023 & 2024                                                                   | Different fiscal years            |
| Chapter 'א (Preamble) & Chapter 'י & Chapter א"י (Commencement) of the Economic Efficiency Law | Different chapters of same law    |
| Chapter 'א & Chapter כ"ו of the Economic Plan Law                                              | Different chapters of same law    |
| צו נכי רדיפות הנאצים & צו נכי המלחמה בנאצים                                                    | Different beneficiary populations |
| Tax ordinance amendment #22 & temporary order #5                                               | Different legislative instruments |
| Ministers' salaries & President's salary                                                       | Different officeholders           |
| Public institution approval lists 2022 & 2023 (section 61)                                     | Different annual lists            |
| Public institution approval lists 2022 & 2023 (section 46)                                     | Different annual lists            |
| 'פרק ט' "except section 36" & "only section 36"                                                | Opposite inclusion scope          |
| Temporary committee chair election & standing committee chair election                         | Different committees              |

Final Statistics

Topic Vocabulary Summary

| Metric                          | Value       |
|---------------------------------|-------------|
| Total protocols                 | 213         |
| Raw unique topic strings        | 180         |
| Genuine typo clusters merged    | 23          |
| False-positive merges corrected | 14          |
| <b>Final canonical topics</b>   | <b>151</b>  |
| Single-topic protocols          | 191 (89.7%) |
| Multi-topic protocols           | 22 (10.3%)  |

Topic Frequency Distribution

| Protocols    | Topics     | Share of topics |
|--------------|------------|-----------------|
| 1            | 121        | 80.1%           |
| 2            | 15         | 9.9%            |
| 3            | 4          | 2.6%            |
| 4            | 3          | 2.0%            |
| 6            | 2          | 1.3%            |
| 7            | 2          | 1.3%            |
| 9            | 1          | 0.7%            |
| 11           | 1          | 0.7%            |
| 12           | 1          | 0.7%            |
| 15           | 1          | 0.7%            |
| <b>Total</b> | <b>151</b> |                 |

**80% of topics appear in only one protocol** (one-off discussions). The remaining **30 recurring topics** (2+ protocols) are the core challenge for streaming subject linking.

### Most Recurring Topics

| #  | Topic                                                            |
|----|------------------------------------------------------------------|
| 15 | פרק ד' (מיסוי בשוק הדיור) --- חוק ההתייעלות הכלכלית              |
| 12 | שינויים בתקציב לשנת 2023                                         |
| 11 | פרק ט' (מיסים) --- חוק ההתייעלות הכלכלית                         |
| 9  | פרק כ' (רשויות מקומיות, סעיפים 78--79) --- חוק ההתייעלות הכלכלית |
| 7  | שינויים בתקציב לשנת 2022                                         |
| 7  | הצעת חוק התקציב לשנת הכספים 2023, מ/1610                         |
| 6  | פרק כ' (רשויות מקומיות, סעיף 79)                                 |
| 6  | הצעת חוק לעידוד תעשייה עתירת ידע, מ/1577                         |

### Implications for Downstream Tasks

- **Topic & segment extraction:** The << >> markers provide near-perfect segment boundary annotations in 96% of protocols. These serve as silver labels and a strong rule-based baseline. The 4 marker-free files require model-based segmentation and are natural test cases.
- **Streaming subject linking evaluation:** Single-occurrence topics are trivially classified as “new”. Evaluation signal comes entirely from the **30 recurring topics**. We recommend sampling the held-out evaluation set exclusively from these, ensuring coverage of both the high-frequency (budget law chapters) and low-frequency (2–4 occurrences) recurring cases.
- **Multi-topic protocols (22 files):** These sessions cover 2–3 distinct legislative matters. Correct segmentation is a strict prerequisite before linking — conflating multiple topics into one embedding degrades recall on the linking task.

**Output files:** `outputs/topics_raw.json`, `outputs/topics_canonical.json`, `outputs/topics_clusters.json`, `outputs/topics_corrections.json`. **Code:** `src/preprocess_topics.py`